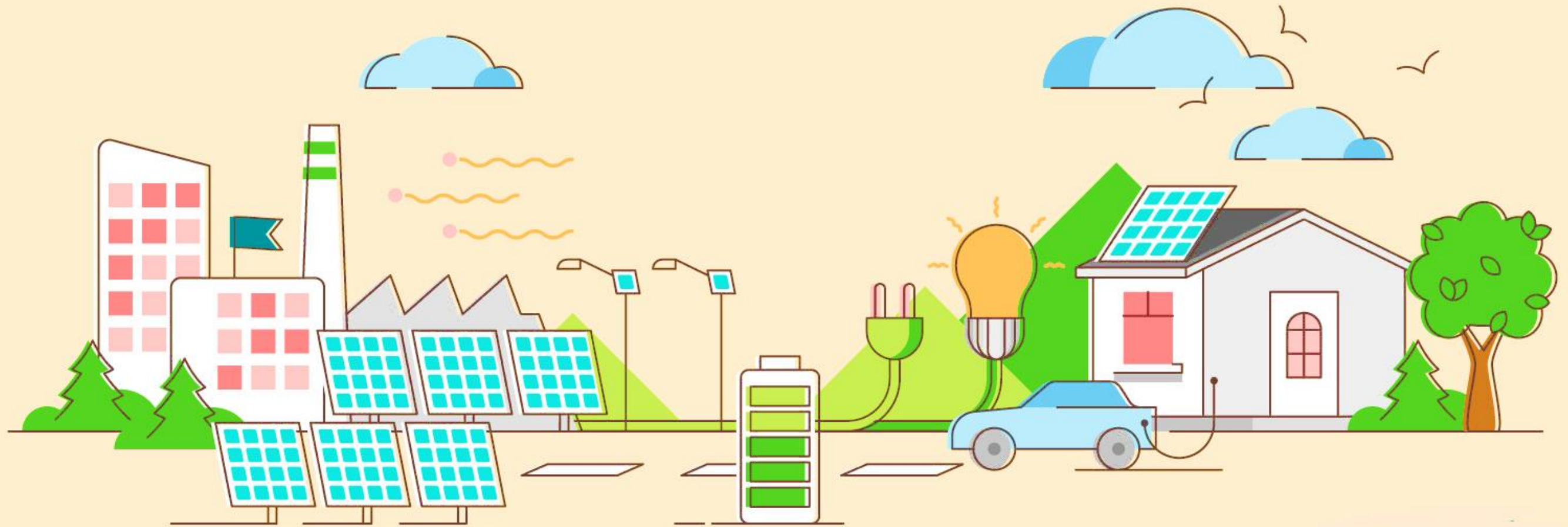
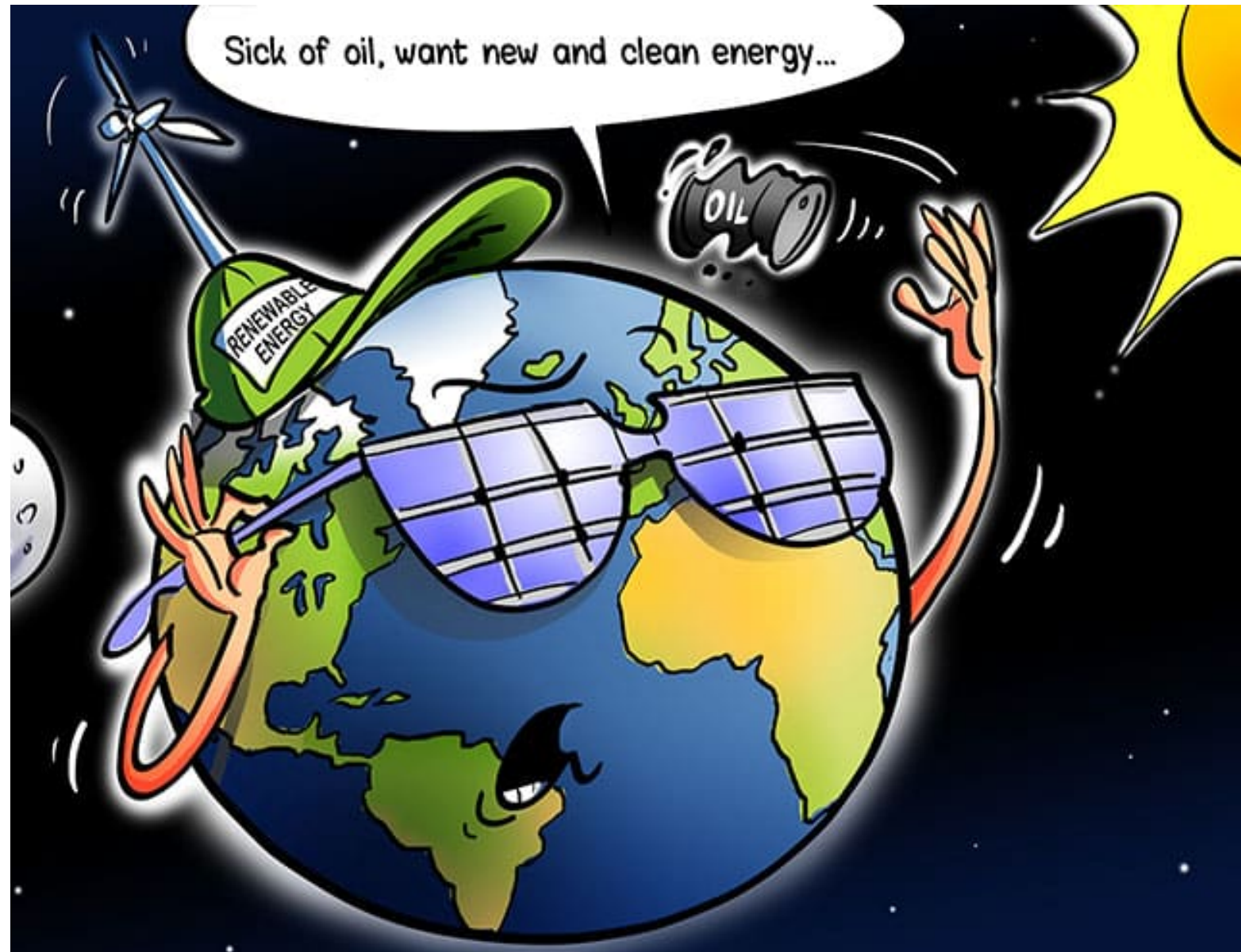


RENEWABLE ENERGY

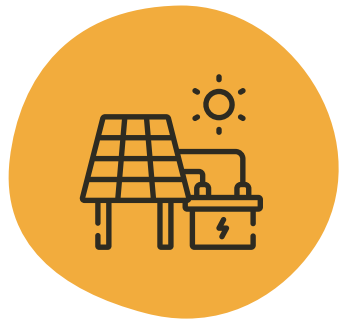


WHAT IS RENEWABLE ENERGY?



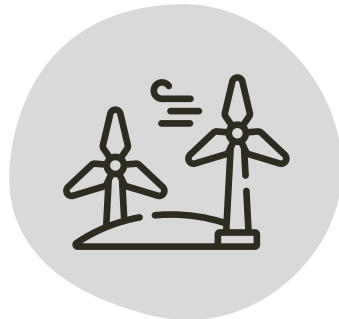
TYPES OF RENEWABLE ENERGY

01



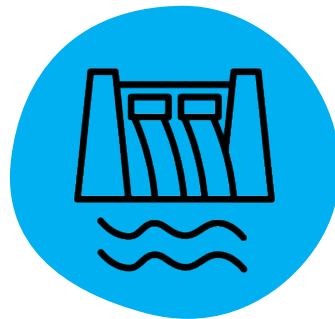
Solar Energy

02



Wind
Energy

03



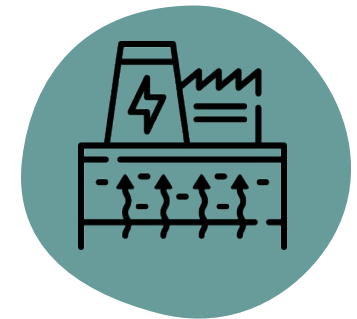
Hydro Energy

04



Biomass
Energy

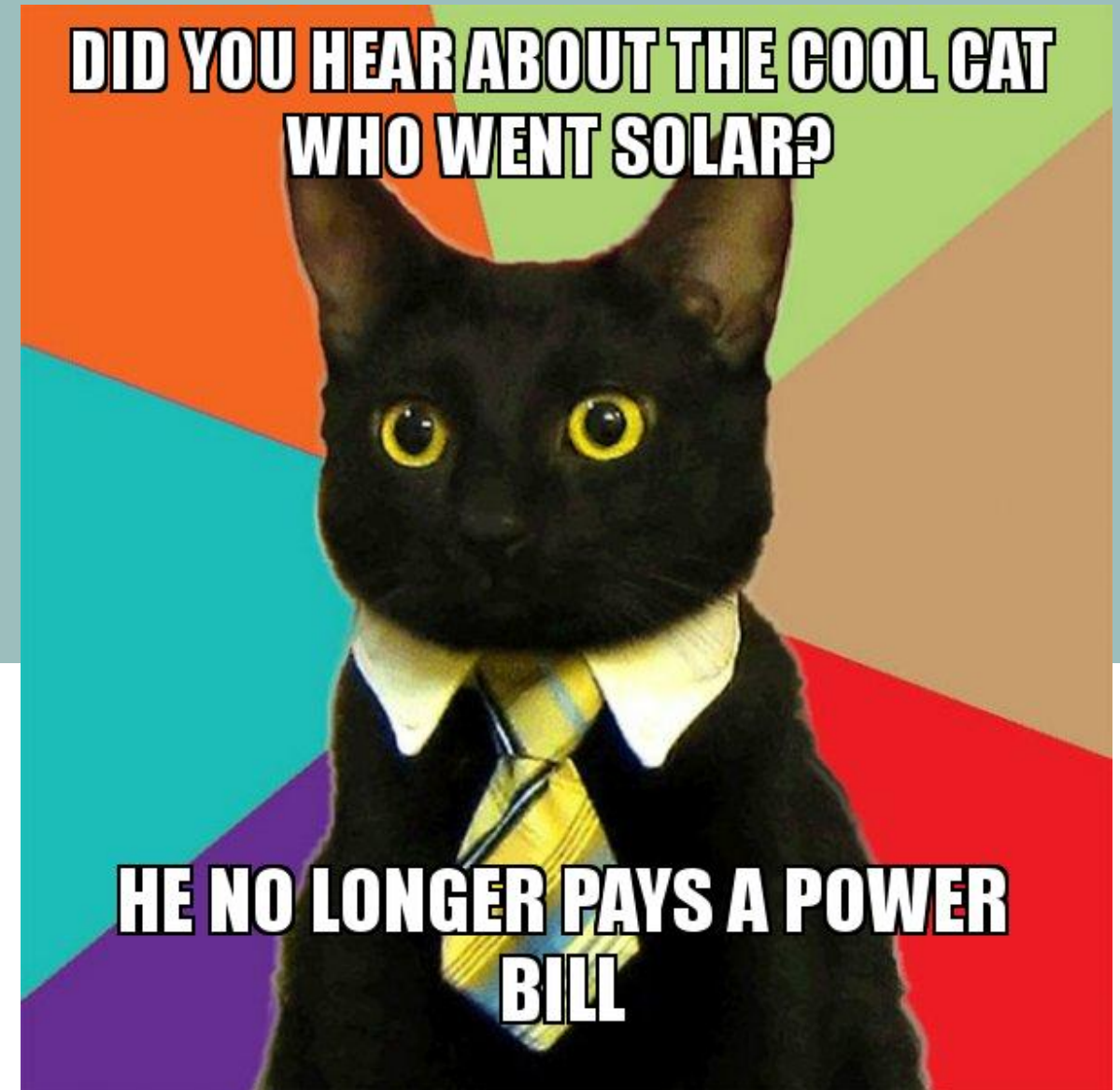
05



Geo Energy

SOLAR ENERGY

Solar power is the conversion of energy from sunlight into electricity, directly using photovoltaic, indirectly using concentrated solar power, or a combination. Photovoltaic cells convert light into an electric current using the photovoltaic effect.



There are two main types of solar energy technologies -

Photovoltaic (PV)

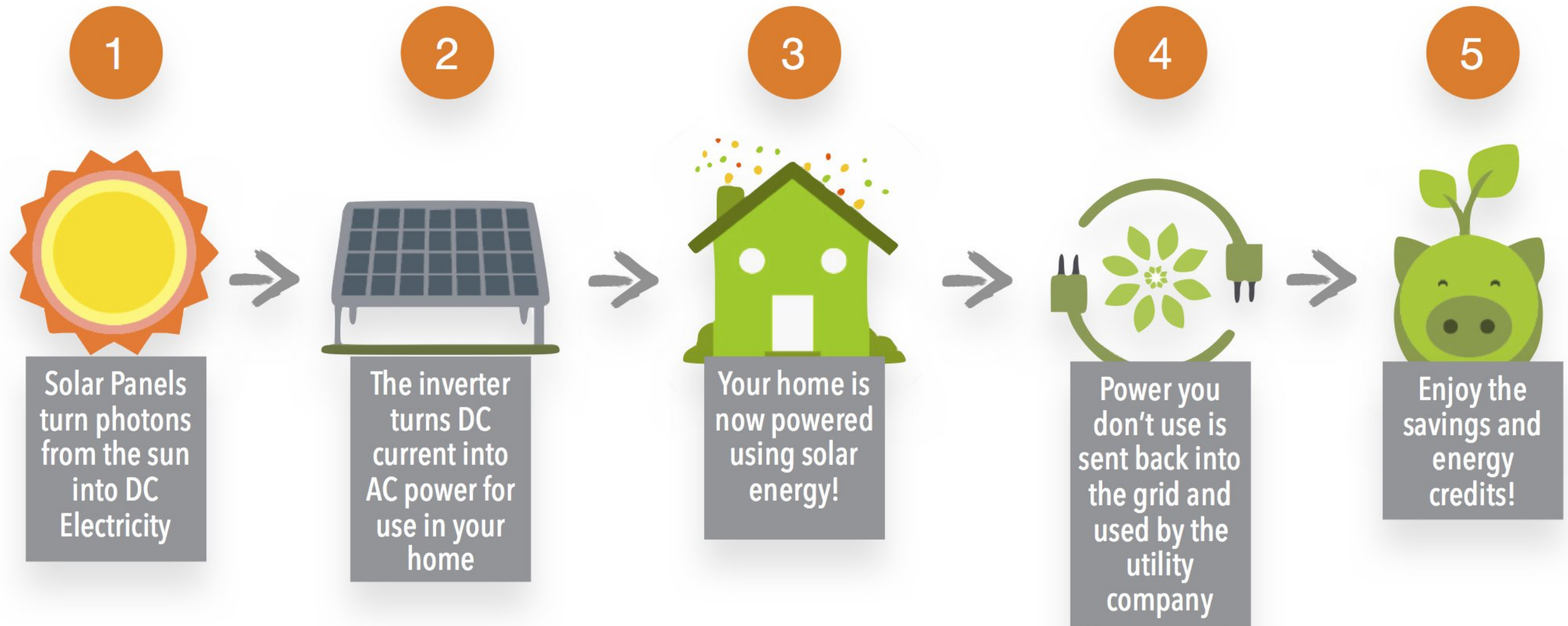
Photovoltaic solar energy is obtained by converting sunlight into electricity using a technology based on the photoelectric effect. It is a renewable, inexhaustible, and non-polluting energy that can be produced in installations ranging from small generators for self-consumption to large photovoltaic plants.



Concentrating solar-thermal power (CSP)

Concentrating solar-thermal power (CSP) technologies can be used to generate electricity by converting energy from sunlight to power a turbine. Still, the same basic technologies can also be used to deliver heat to a variety of industrial applications, like water desalination, enhanced oil recovery, food processing, chemical production, and mineral processing.





Benefits of Solar Energy-

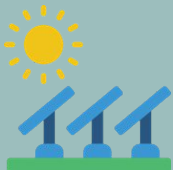
Impact on the Environment



Reduce Your Energy Bill



Solar Energy Is Applicable Everywhere



Less Electricity Loss During Long-Distance Transport



WIND ENERGY

km or more from land).

Wind power or wind energy describes the process by which the wind is used to generate mechanical power or electricity.

They are usually grouped in wind farms (sometimes called wind parks). There are onshore farms (which, however, are often near water); near-shore farms (on land or sea within several km of coast); and offshore parks (ten



Tauern wind park,
Austria

Onshore

Topography disrupts
wind flow



Offshore

Requires shallow seabed of
about 50m to be mounted



Floating

Faster ocean winds generate up
to eight times as much power

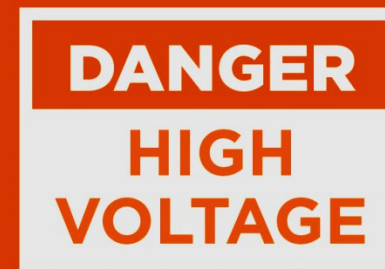




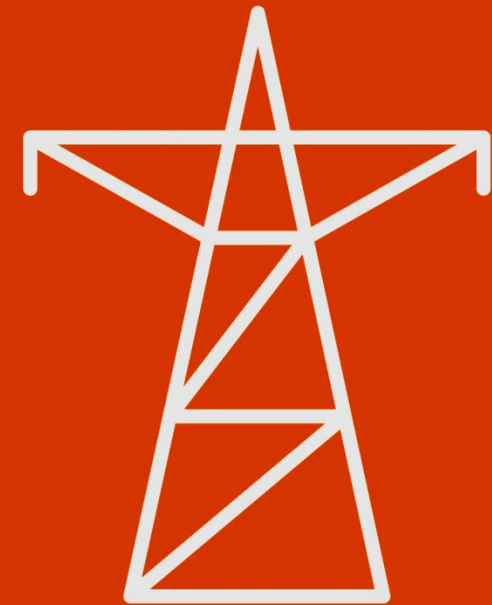
**Wind
Turbine**



Transformer



Substation



**Energy
Grid**

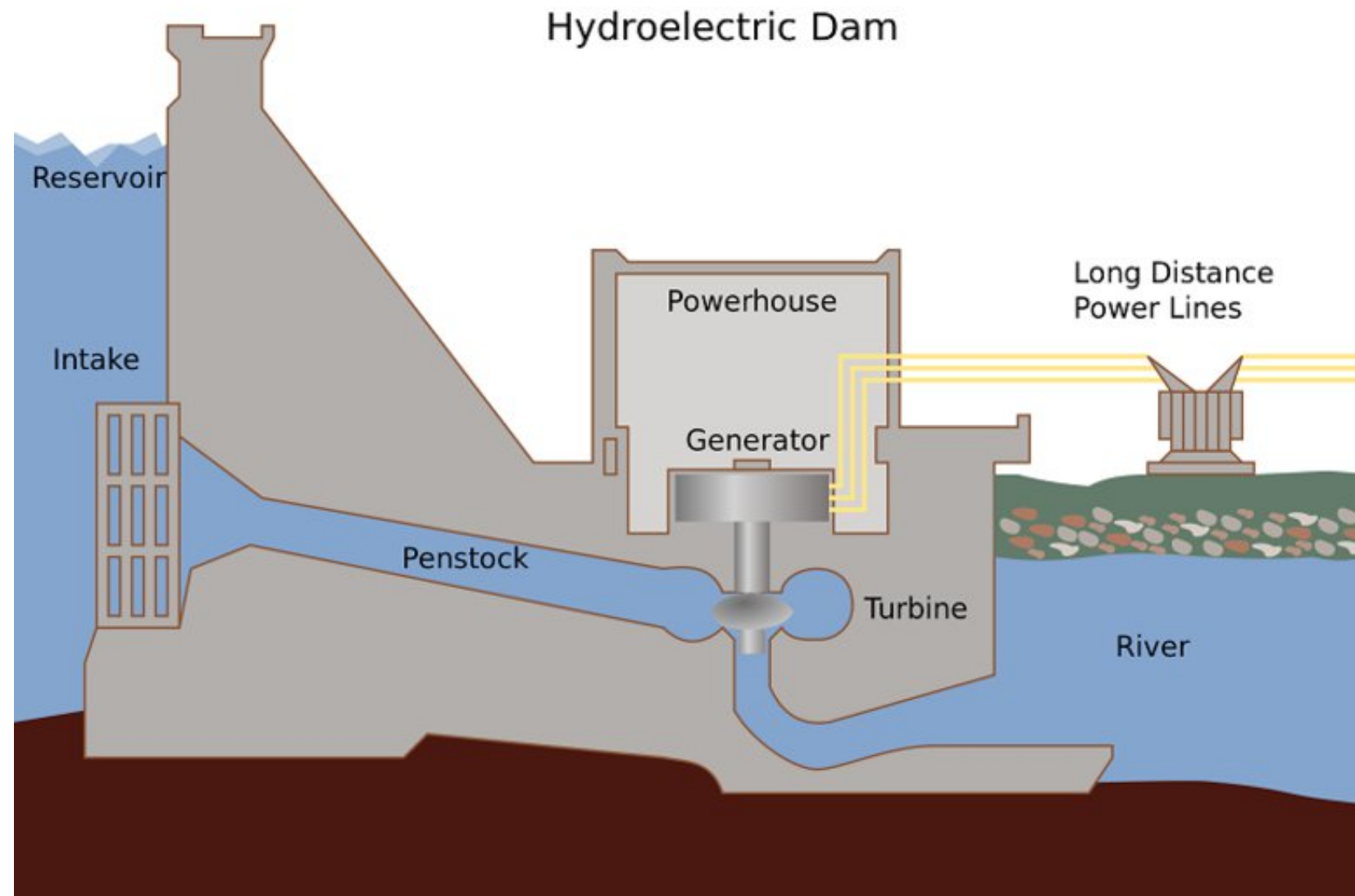
HYDRO ENERGY

Hydropower, or hydroelectric power, is a renewable energy source that uses a dam or diversion structure to alter the natural flow of a river or other body of water.



Øvre Forsland

HOW HYDRO ENERGY WORKS?



Perks of hydro energy-

1. The cheapest energy source

Even though the initial investment required to build a plant is very high and something of a challenge, waterpower is the cheapest source overall in the medium to long term. Once the dams and stations have been built and the turbines installed, they require only minimally expensive maintenance compared to the initial investment.

2. Hydropower stations are agile and responsive

Faced with water availability that can change over time, the power stations are extremely flexible. Production systems require a very small amount of energy to start working.



3. Water brings with it enormous energy

The energy potential of hydropower is enormous. The gigantic masses of water found at high altitudes have a significant amount of gravitational potential energy and even using just a part of it will provide abundant power.

4. Production can be tailored to demand

The water flow can be easily controlled to suit energy requirements. This is true both in the short term - for example, in the night-day cycle - and in the longer term for seasonal variations and periods of drought.



5. Hydropower as a tool for reclamation...

With water courses contained within pre-established reservoirs, hydropower stations also help avoid flooding as well as helping to drain marshy land where stagnant water tends to accumulate. The flow and total water volume can be controlled very precisely. Thus, it is possible to ensure that the water flows steadily, even where rainfall tends to be intense and concentrated in short periods.

6. ... and as an irrigation system

At the opposite end of the spectrum to the previous benefit, hydropower stations can also prove invaluable where water is very scarce. Reservoirs can be used as water reserves for periods of drought, in part because the water coming out of the station is perfectly clean and usable. The opening of dams can also be planned and calibrated around crop cycles and ecosystems, and the water can be allowed to flow to allow fish get through at appropriate times.



BIOMASS ENERGY



**LANDFILL
GAS**



GARBAGE



**ALCOHOL
FUELS**

TYPES OF BIOMASS



WOOD



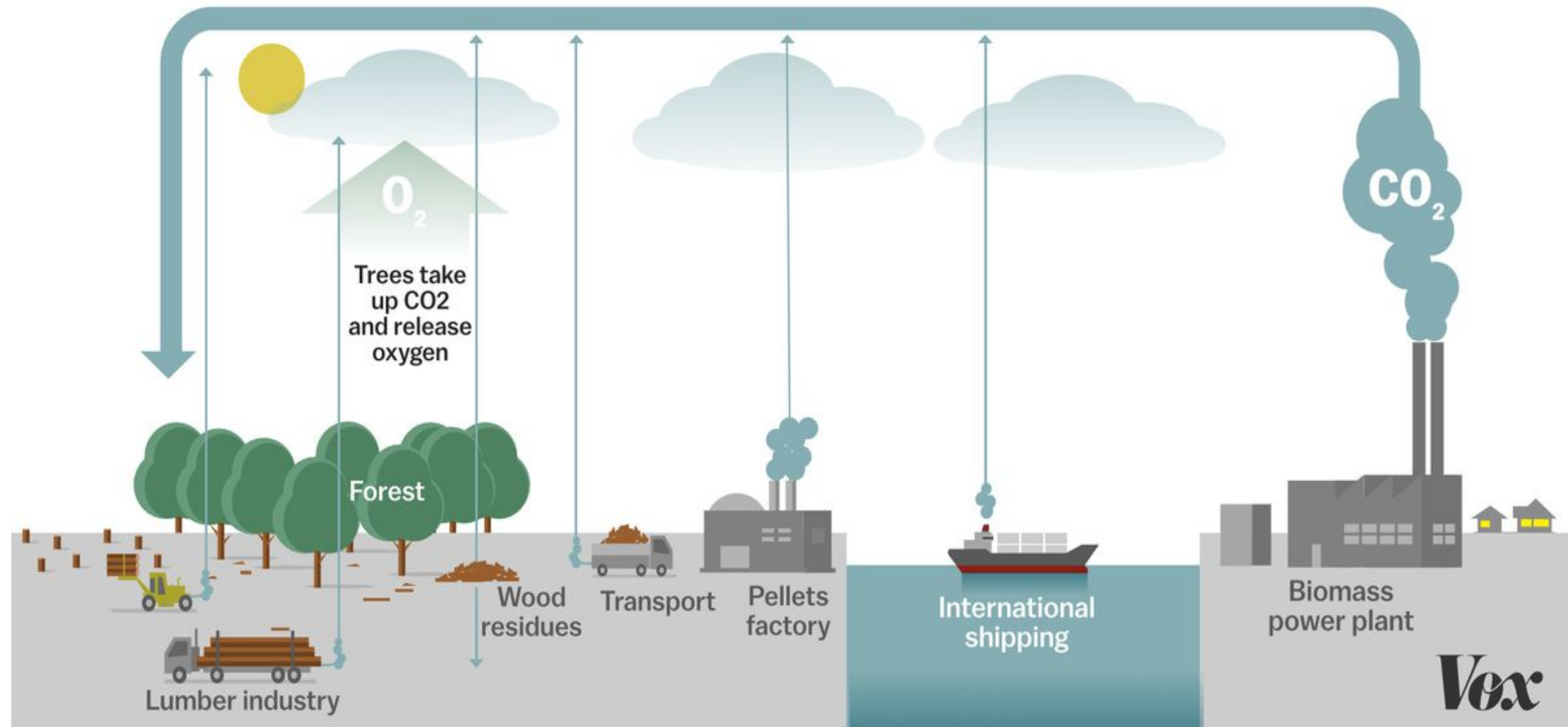
CROPS

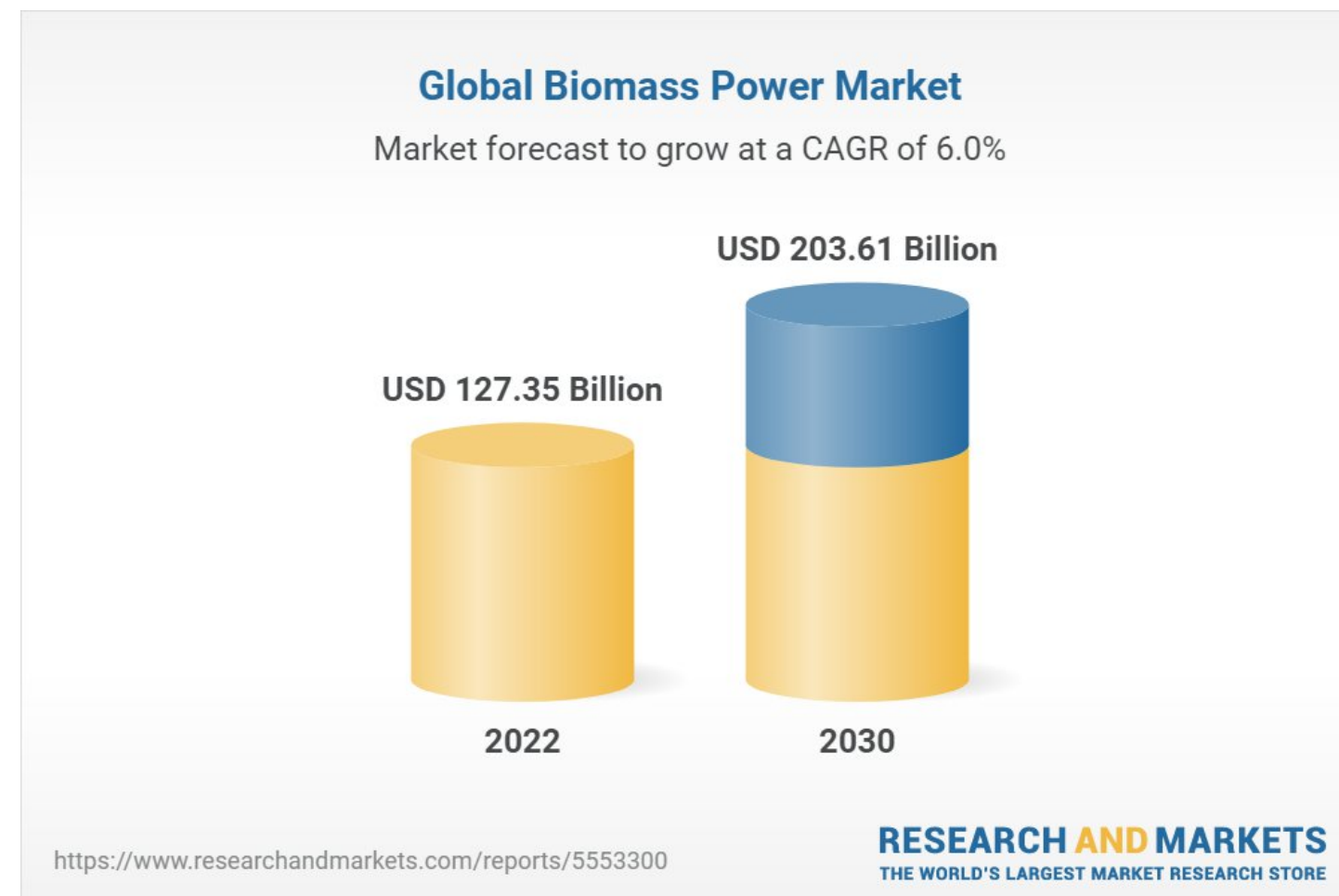
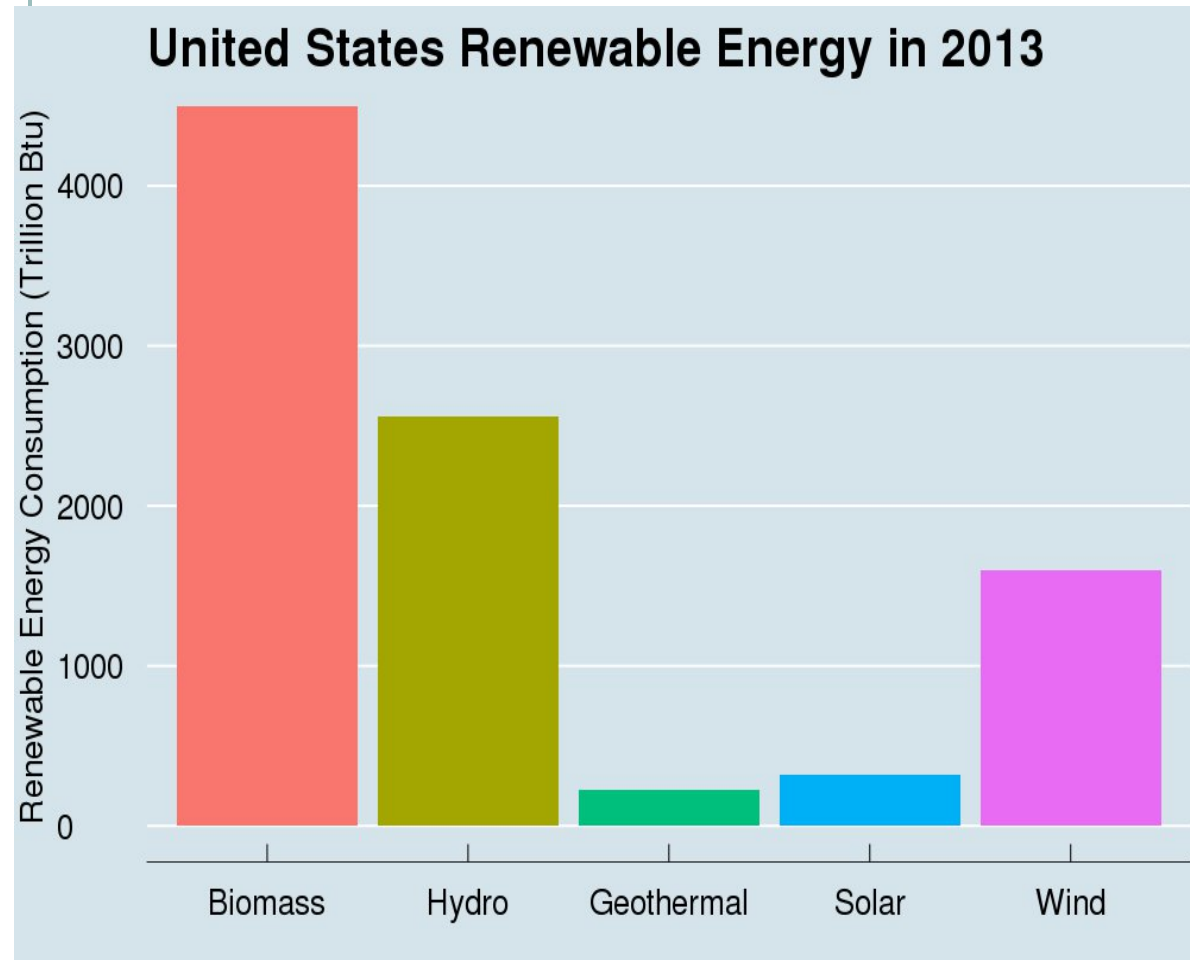
Biomass is a plant-based material used as fuel to produce heat or electricity. Examples are wood and wood residues, energy crops, agricultural residues, and waste from industry, farms, and households. Since biomass can be used as a fuel directly, some people use the words biomass and biofuel interchangeably.

Biomass is probably our oldest source of energy after the sun.



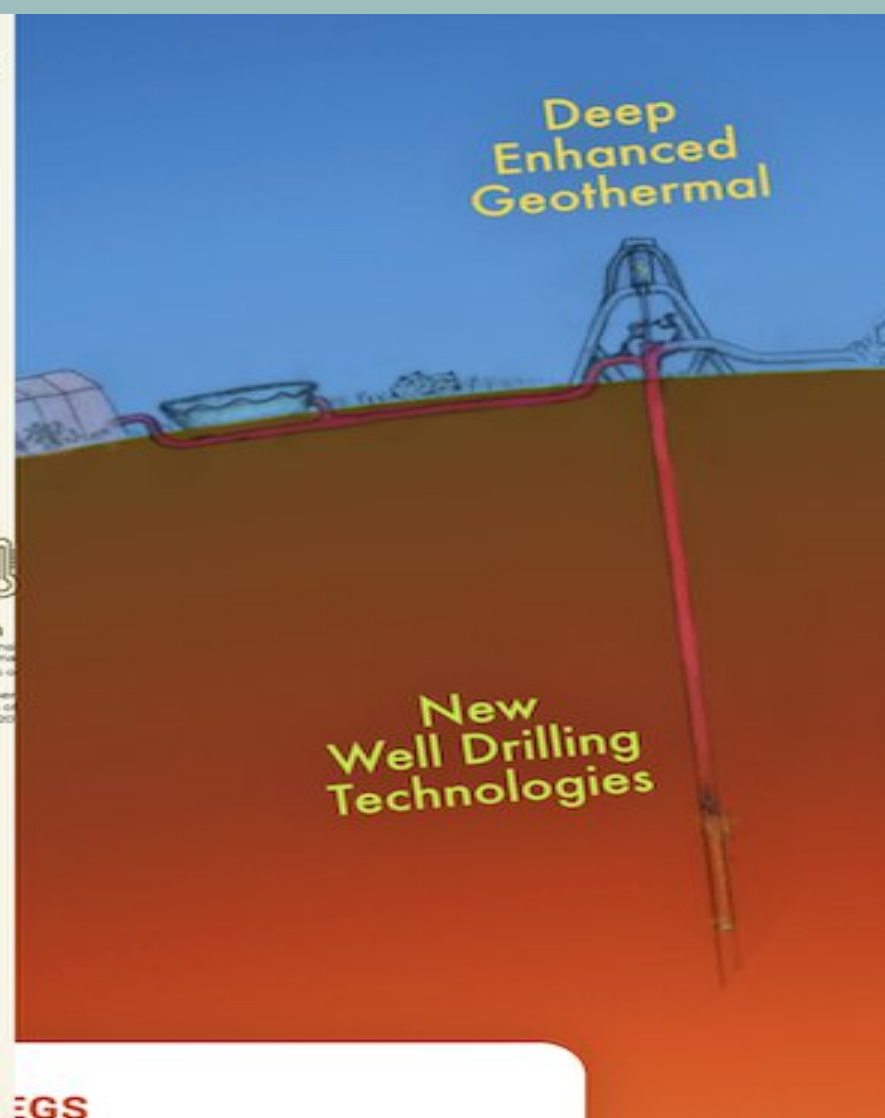
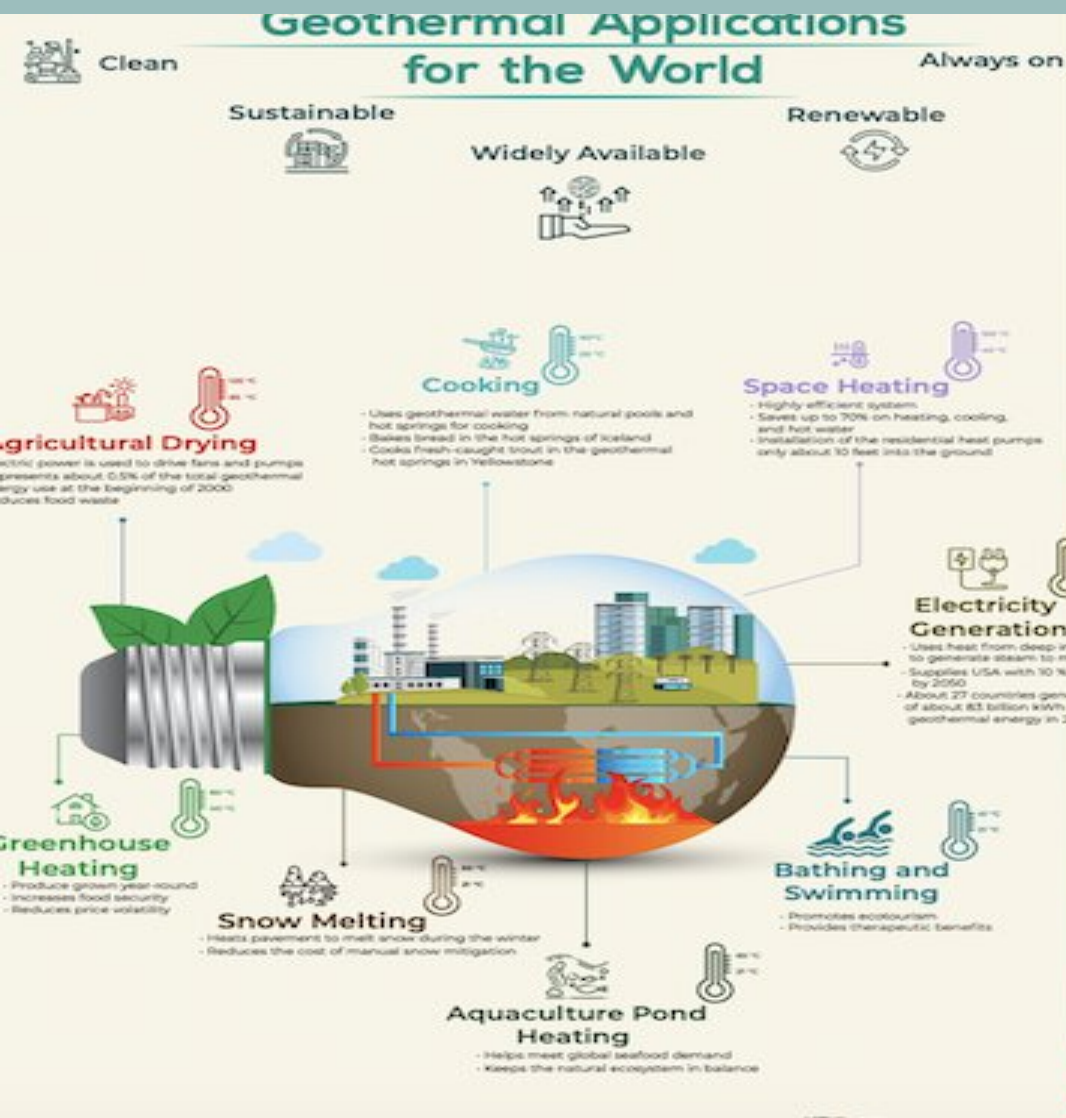
The cycle of biomass energy





GEO ENERGY

Geothermal energy is heat that is generated within the Earth. (Geo means “earth,” and thermal means “heat” in Greek.) It is a renewable resource that can be harvested for human use. About 2,900 kilometers (1,800 miles) below the Earth's crust, or surface, is the hottest part of our planet: the core.



THE HEAT BENEATH OUR FEET

AVC ENVIRONMENTAL

When looking for a carbon-neutral way to generate heat in our homes, Geothermal Heating Pumps (GHPs) are an effective solution.

HEATING

- A geothermal heat pump can use underground heat to send warm air through a system.
- Plastic pipe is laid in a loop beneath the ground outside of the system either flat or running straight down.
- Brine, water, or antifreeze solution is pumped through the pipe, which picks up the underground heat.
- The heat pump takes heat from the solution and uses it to pump warm air through tubes throughout the system, this process is called “heat exchange”.

GEO HEAT PUMPS

Provide savings of up to **72%** on heating & cooling costs.

Most small island economies rely on oil-fired power plants to provide a steady electricity supply, but **Iceland** has virtually 100% renewable electricity from its abundant hydropower and geothermal resources.



Nesjavallavirkjun

Perks of Geo Energy-

- 1. Baseload energy - it's always on:** Geothermal power plants produce electricity consistently, running 24 hours a day, 7 days a week. The power output of a geothermal power plant is highly predictable and stable, thus facilitating energy planning with remarkable accuracy.
- 2. Geothermal energy can heat, cool, and generate electricity:** Geothermal energy can be used in different ways depending on the resource and technology chosen—heating and cooling buildings through geothermal heat pumps, generating electricity through geothermal power plants, and heating structures through direct-use applications.

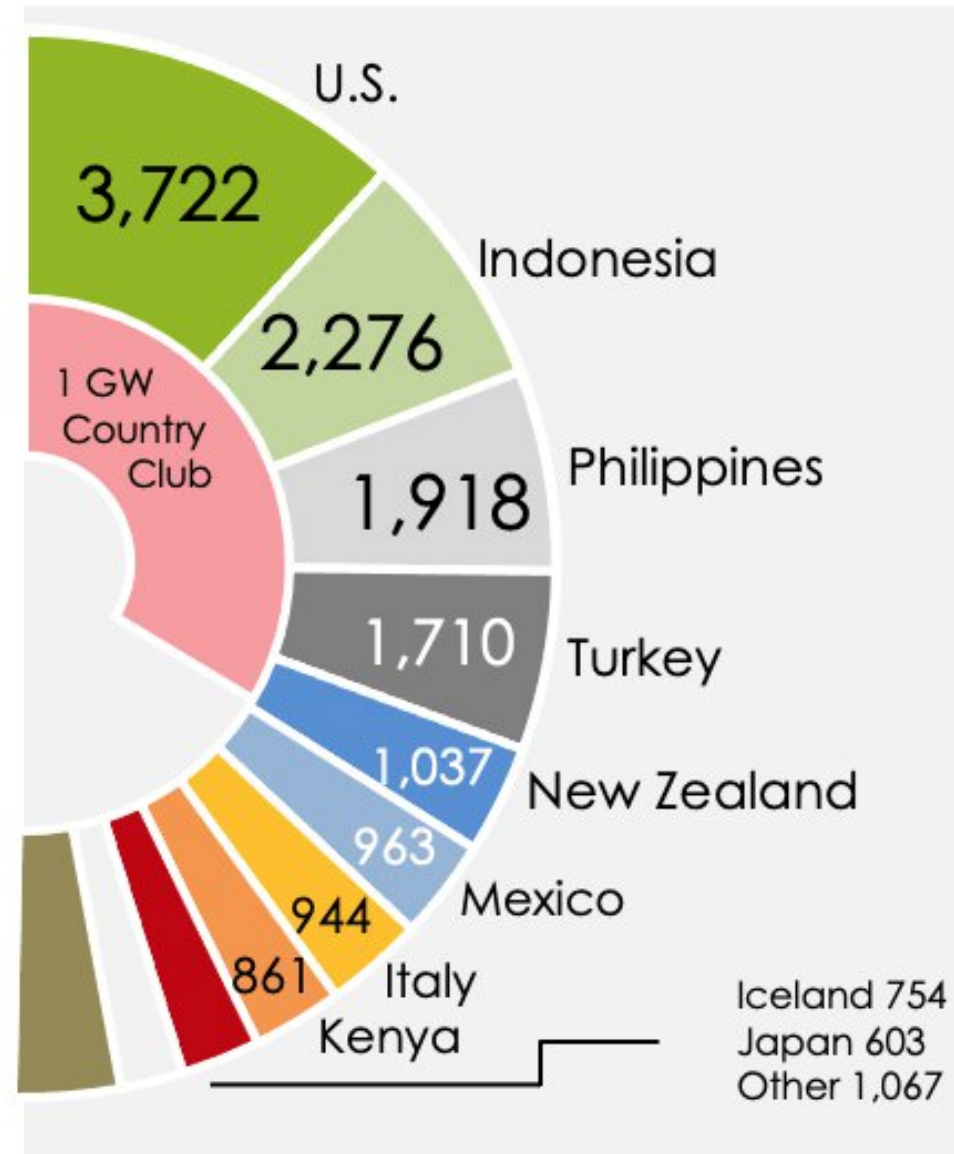
3. Over 100 years of geothermal sourced energy: The first ever geothermal plant was set up in Larderello, Italy 1904. Steam from that geothermal source was used to turn a small turbine that powered five light bulbs. Today, the U.S. leads the world in geothermal power generation, providing more than 3.7 GW to the national grid.

4. Weather Independence: A significant advantage is that geothermal energy is not dependent on weather conditions.

Top 10 Geothermal Countries 2021

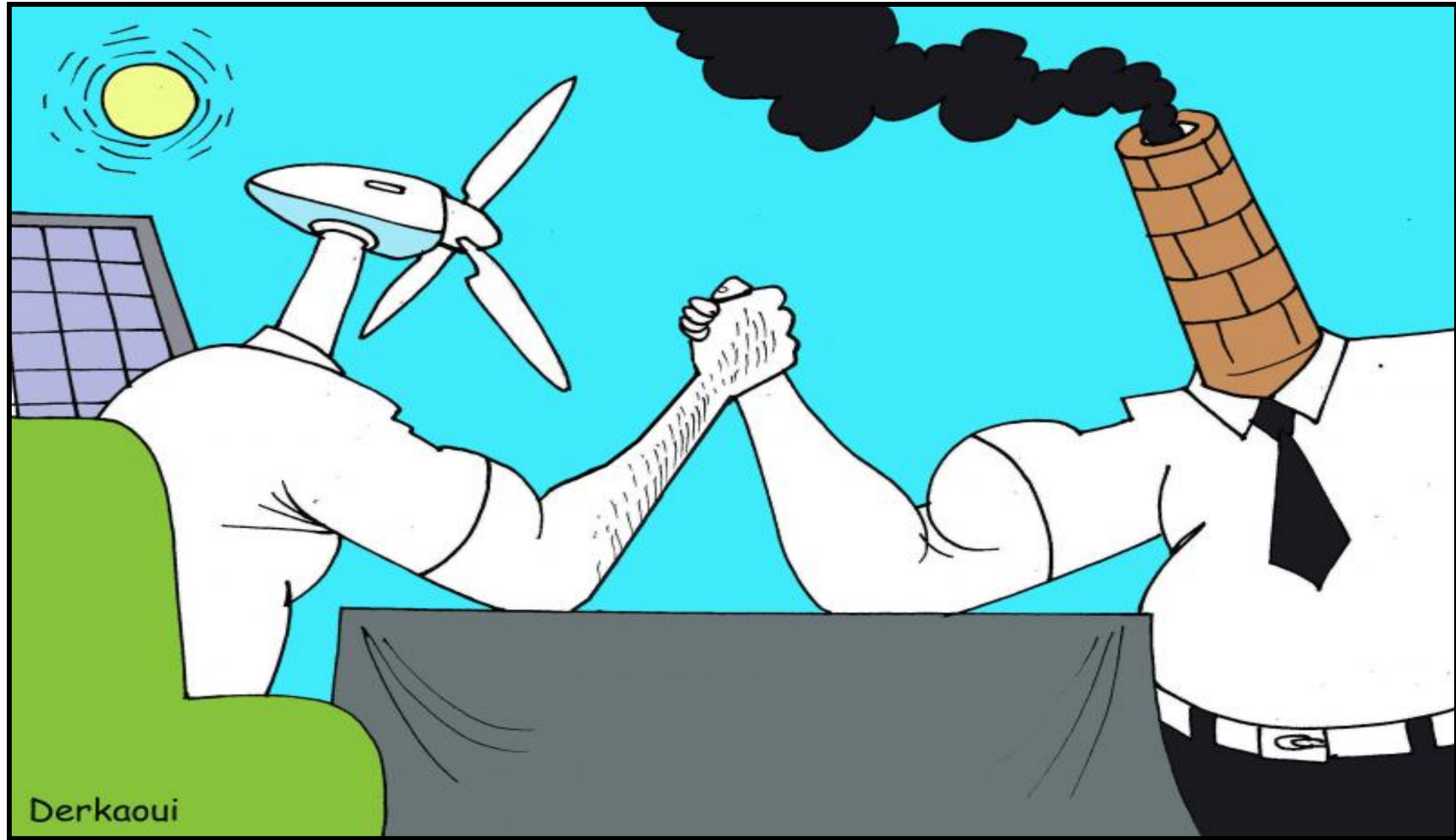
Installed Capacity in MWe
Year-End 2021

Total 15,854 MW



Source: ThinkGeoEnergy Research 2022

WHO WILL WIN THIS BATTLE ?



BENEFITS OF RENEWABLE ENERGY

☐ **46M** pounds of annual reductions
in GHG emissions

☐ **10M** gallons in annual water
savings

☐ **225** acres of land preserved by
using roofs and parking lots



BENEFITS OF RENEWABLE ENERGY

RENEWABLE ENERGY BENEFITS: MEASURING THE ECONOMICS



ADVANTAGES & DISADVANTAGES OF RENEWABLE ENERGY

- ❑ Easily regenerated
- ❑ Supports environment
- ❑ Easily available
- ❑ Low maintenance cost
- ❑ Weather dependency
- ❑ High installation cost
- ❑ Fluctuation problem(Solar)
- ❑ Geographic limitations

**YOU DON'T LIKE RENEWABLE
ENERGY ?**



COAL STORY BRO

makeameme.org

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- <https://www.youtube.com/watch?v=PROhNDen3nk>